

Bloculus - Project Brief

Cedwyn Estate Pilot

Version 1.0.0 · 1 November 2026

Author(s): DATRO Consortium

Copyleft 2021-2026, DATRO Consortium. Earlier editions are listed in the Release Notes.

Bloculus - Project Brief: Cedwyn Estate Pilot

Bloculus pairs each place on the ground with one stable Uber H3 hexagon reference, and gives that reference its own web address. This edition (version 1.0.0) makes the Cedwyn estate, Llandough and Leckwith, south-east of the river Ely in south Wales, the first Bloculus pilot. Each of the estate's 59 numbered land parcels is proposed to carry the H3 cell of its centre point, and that cell its own sub-domain under `bucklerfamily.estate`.

Release Notes and Notices

This section provides information about what is new or changed, including urgent issues, documentation updates, maintenance, and new releases. - 'Updates' are the term used to describe significant changes to our public source code and/or records.

This Release (Version 1.0.0)

- **2026-Nov-01** - Major release: the brief is retargeted to the Cedwyn estate (Llandough and Leckwith, south-east of the river Ely) as the first Bloculus pilot. The earlier coastal pilot area is retired from this document.
- **2026-Nov-01** - All addresses move to bucklerfamily.estate (sub-domains such as <h3 cell>.cedwyn.bucklerfamily.estate) until a dedicated top-level domain, for example .cedwyn, is available.
- **2026-Nov-01** - Kept the H3 thesis (one hexagon reference per place, addressable, routable). Added the Cedwyn specifics: 59 numbered land parcels, one H3 cell per parcel centre point, and a parcel sub-domain proposal.
- **2026-Nov-01** - Sale prices, investment figures and royalty examples from the earlier edition are removed; the pilot's model is a proposal with no prices set.
- **2026-Nov-01** - New annex: H3 layers over the pilot area (resolutions 8, 9 and 11). The earlier coastal annex maps are retired.
- **2026-Nov-01** - Built in English. German, Spanish and French editions of 1.0.0 are pending translation (see Known Issues).

Older Versions

In the table below each entry links to an archived copy of an earlier edition. The date the file was archived will differ from the date of the document label, which is its creation date.

Older Versions of this Document

Archive Date	Version	Description	Download Link
2021-08-24	0.0.0	see releasenotes	2021-08-24_v0.0.0.pdf
2026-11-01	0.0.1	earlier coastal pilot edition	v0.0.1 PDF (en)

Version 0.0.1

- **2021-Aug-24** - Archived version 0.0.0 and updated the latest release to semantic version 0.0.1
- **2021-Aug-24** - Rebuilt using new rebuild.sh script (adds copyleft date range in footer)
- **2021-Aug-24** - Added some more text - mainly to justification segment

Version 0.0.0

- **2021-Jul-24** - Added SVG download links in Annexures
- **2021-Jul-11** - Added Annexures with Resolutions 5 and 6
- **2021-Jul-09** - Document title changed to “Business Case”
- **2021-Jun-24** - Changed document title to match directory path

Known and Corrected Issues

Below is a table of pending issues which have been reported to our team. When these issues are remedied, or any significant changed made, a new release will be published.

Known Issues

Date	Version	Subject	Description
2021	0.0.0	security	Having a node’s H3 cell reference in HotspotBnB before it self-builds is useful but the solution must circumvent bad actors
2026-11-01	1.0.0	translation	The German / Spanish / French editions of 1.0.0 are pending. The 0.0.1 translations describe the retired coastal pilot

Date	Version	Subject	Description
2026-11-01	1.0.0	geometry	Parcel centre points and their H3 cells are approximate until the 59-parcel list is traced over modern mapping. Each cell will name its source and tolerance
2026-11-01	1.0.0	addressing	Two-level sub-domains (<cell>.cedwyn.bucklerfamily.estate) need a certificate that covers *.cedwyn.bucklerfamily.estate before they can serve HTTPS

Executive Summary

The Cedwyn estate lies in Llandough and Leckwith, south-east of the river Ely, just outside Cardiff in south Wales. Its record and chronology are kept at cedwyn.bucklerfamily.estate, which organises the estate as 59 numbered land parcels, each with its own webapp (shown as “NN of 59”). This brief proposes the Cedwyn estate as the first pilot of the Bloculus protocol: every parcel’s centre point is tied to one H3 hexagon reference, and that reference becomes the parcel’s address on the web.

Bloculus began in 2021 as a component of DATRO’s community network work. This edition keeps its central idea, an addressing system in which the address itself carries the physical location, and applies it to a real, bounded, well-documented piece of land. The DATRO source code remains public under a General Public Licence, so anyone can run the same approach for their own place.

Statements about the estate in this brief are limited to what the Cedwyn site publishes. Everything else in it, the cell choices, the addresses and the pilot plan, is a proposal.

Challenge

The pilot sets out to solve five problems:

1. Every parcel, and every thing pinned to a parcel (a tour stop, a reconstruction, an archive record, a game object), needs one stable reference that is the same for people, maps and machines.
2. That reference must resolve to the physical ground, so a phone standing in a field can find the content for that field, the way location games pin characters to real places.
3. References must be short enough to travel over low-bandwidth links and to be typed, printed on a sign or encoded in a QR code.

4. The same reference must group naturally into larger areas (parcel, farm, estate, parish) without a second system.
5. The layout must be portable: the planned replica of the estate in Patagonia should be able to re-use the parcel layout while keeping its own real-world references.

Solution & Outcomes

The protocol is named Bloculus, a combination of two words:

- ‘Bloc’, a reference to the blocks of a ledger and to blocks of land
- ‘Loculus’, an architectural and biological word for a “little place” or compartment

Bloculus combines Uber’s hexagonal (“H3”) grid reference system with the domain name system. The result is a web address for each hexagon’s unique reference. Addresses work first on a local network (on site, without internet), and resolve on the wider internet once a gateway exists.

There are many grid reference systems, for example the Ordnance Survey National Grid, Universal Transverse Mercator (UTM) and the Military Grid Reference System (MGRS). H3 is unusual: instead of a point or a square it uses hexagons, often called “honeycombs”. Each hexagon has neighbours at equal distances, and each one splits into about seven smaller ones, so one family of references runs from continents down to a few square metres. H3’s “resolution” sets the honeycomb size. The resolutions that matter for Cedwyn are:

H3 Resolution	Avg. Hexagon Area	Avg. Edge Length	Role in the Cedwyn pilot
8	0.737 km2	531 m	estate and village scale
9	0.105 km2	201 m	farm and field groups
10	15,048 m2	76 m	large parcels
11	2,150 m2	29 m	parcel centre point (proposed)
12	307 m2	11 m	buildings, tour stops, pins

Figures are H3’s published averages (H3 library 4.x). Each reference is a 15-character code. For example, near the centre of Llandough village (an approximate point, within about 300 m) the references are:

- resolution 8: `8819586b39fffff`

- resolution 9: `8919586b393ffff`
- resolution 11: `8b19586b3923fff`

The 2021 edition chose resolution 8 (about 500 m honeycombs) to balance privacy and accuracy for network equipment rooms. A land pilot needs finer cells. This brief proposes **resolution 11 for each parcel's centre point**: a hexagon of about 2,150 m2, small enough to sit inside a typical field, large enough to stay stable when the traced centre point moves by a few metres. Resolution 12 is kept for things pinned inside a parcel, such as a tour stop or a reconstruction anchor. Because H3 is hierarchical, every parcel cell also names its resolution 9 and resolution 8 parents, so the farm and the estate fall out of the same codes.

Use of H3 references cannot be owned or controlled by anyone; they are a public grid. What can be held is the name that points to them. The 2021 edition proposed one Handshake top-level domain per hexagon (for example `.8843a13687fffff`). Until a suitable top-level domain exists for the estate (for example `.cedwyn`), the pilot uses sub-domains of `bucklerfamily.estate`:

- parcel: `<resolution 11 cell>.cedwyn.bucklerfamily.estate`, with a readable alias `p01.cedwyn.bucklerfamily.estate` to `p59.cedwyn.bucklerfamily.estate`
- a service or layer on a parcel: `<service>.<cell>.cedwyn.bucklerfamily.estate`, for example `tour.<cell>.cedwyn.bucklerfamily.estate` or `archive.<cell>.cedwyn.bucklerfamily.estate`
- later, with a dedicated top-level domain: `<cell>.cedwyn` and `tour.<cell>.cedwyn`

These names are a proposal. None is registered or live yet; each will be added only when its parcel's centre point has a named source.

Because the address contains the location, a device on site can work out which parcel it stands in, look up that parcel's cell, and open its content directly; services in neighbouring parcels can find each other by asking for the neighbouring cells. A query for content in one place can start at the estate cell and narrow down through its children, the same triangulation idea the 2021 edition described for routing.

Pilot Model (proposal)

The 2021 edition priced honeycombs for sale and modelled investor returns. This edition sets no prices and makes no financial projections. The pilot is a public reference layer for one estate:

- one record per parcel: parcel number (1 to 59), name, webapp, centre point (latitude and longitude), H3 cell at resolution 11, the source of the centre point and its tolerance
- the 2D and 3D maps on the Cedwyn site draw parcels from that record, so the maps, the webapps and the addresses agree

- content in each webapp (tour notes, records, reconstructions, games) is pinned to a cell, so it appears in the right place on the ground and on the map
- the planned replica of the estate in Patagonia re-uses the parcel layout (parcel numbers and relative positions); its own cells are computed from its own planned site, and it is always labelled planned

Recommendation

1. Trace the 59 numbered parcels from the historic estate plan over modern mapping, record each centre point with its source and tolerance, and compute its resolution 11 cell.
2. Publish the parcel record as one data file on the Cedwyn site, and draw the 2D and 3D estate maps from it.
3. Propose one sub-domain per parcel under `cedwyn.bucklerfamily.estate` and add them only once each centre point is sourced; arrange a certificate covering the second level first.
4. Pin site content to cells (resolution 11 for parcels, resolution 12 for points inside them).
5. Review the top-level domain question (for example `.cedwyn`) once the pilot addresses are in use.

Justification

A bounded estate with numbered parcels is the right size to prove the protocol: small enough to finish, rich enough to test every layer (land, buildings, routes, records, reconstructions and an overseas replica). H3 gives one public, hierarchical reference that maps, phones and software all read the same way, and it does not change when a map is redrawn. Putting the reference in the web address means a person, a printed sign and a machine all reach the same page for the same piece of ground.

The wider DATRO vision is unchanged: services hosted locally, reachable by place, working on a local network first and joining the internet through a gateway. HotspotBnB, DATRO’s self-building localhost web server, can carry a parcel’s cell in its configuration so that a device placed on that parcel identifies itself and pairs with its neighbours.

Supporting Material

An annexure is a supplement to this document; see the Annexures chapter.

Finance

No budget is set for the pilot and no prices are proposed. The parcel record, the maps and the parcel pages run on the existing static hosting of the Cedwyn site. Any cost that

becomes necessary (for example a certificate covering second-level sub-domains, or a dedicated top-level domain) will be listed here as a proposal before it is incurred.

Project Definition

Background information

Bloculus was first written in 2021 for a coastal community network. This edition moves the pilot to the Cedwyn estate (Llandough and Leckwith, south-east of the river Ely), whose record is kept at cedwyn.bucklerfamily.estate.

Business Objective

One stable, public, location-bearing reference and web address for each of the estate's 59 numbered land parcels.

Benefits and Limitations

Benefits: maps, webapps, signs and devices share one reference per parcel; content can be pinned to the ground; the layout can be re-used for the planned replica in Patagonia. Limitations: H3 hexagons do not follow field boundaries, so a cell marks a parcel's centre, not its shape; parcel geometry is approximate until traced and sourced.

Option Identification & Selection

Options considered: postcodes (too coarse, not stable for open land), National Grid references (square, not hierarchical by area), What3Words-style words (proprietary), H3 (public, hierarchical, equal-neighbour hexagons). Selected: H3, resolution 11 for parcel centres (proposal).

Scope, Impact, and interdependencies

In scope: the 59 parcels, their centre points and cells, the parcel record, and the sub-domain proposal. Depends on the estate plan being traced over modern mapping and on the Cedwyn site's 2D and 3D maps.

Outline Plan

1. Trace parcels and record centre points with sources and tolerances. 2. Compute cells and publish the parcel record. 3. Draw the estate maps from the record. 4. Propose and then add parcel sub-domains. 5. Pin content to cells.

Market Assessment

Not assessed for this pilot; no prices are proposed.

Risk Assessment

Approximate geometry (mitigation: every cell names its source and tolerance); certificate coverage for two-level sub-domains (mitigation: arrange before going live); privacy (mitigation: publish only what the Cedwyn site already shows).

Project Approach

Small, verifiable steps on the existing static sites; each step published as a new semantic version of this brief.

Purchasing Strategy

None. Any purchase would first be listed as a proposal under Finance.

Project Organization

Project Governance

The DATRO Consortium maintains this brief. Changes are released as new semantic versions with release notes.

Progress Reporting

Progress is recorded in the release notes of each edition and in the parcel record’s own change log on the Cedwyn site.

Annexures

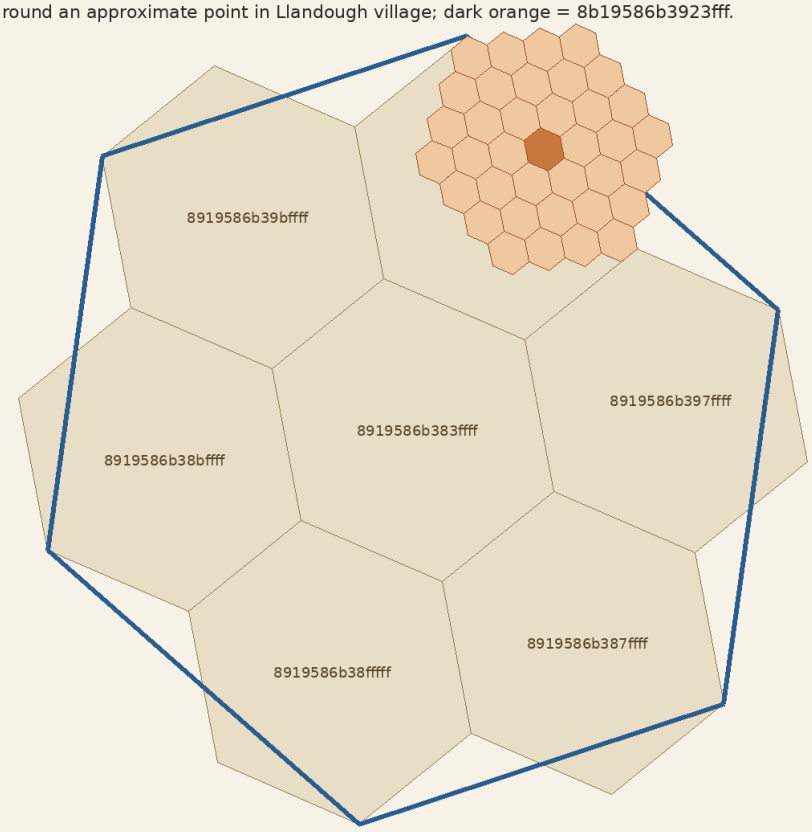
This section contains the Annexures referenced in the text above. The Download Links for these Annexures can be found at the very end of this document.

H3 layers over the pilot area (Resolutions 8, 9 and 11)

A schematic of the H3 hierarchy around an approximate point near the centre of Llandough village: the resolution 8 cell (about 531 m edge), its resolution 9 children (about 201 m edge) and the resolution 11 cells (about 29 m edge) proposed for parcel centre points. Parcel outlines are not drawn; they will be added once the 59 parcels are traced and sourced.

Bloculus, Cedwyn estate pilot: H3 resolutions 8, 9 and 11 (schematic)

Blue outline: res 8 cell 8819586b39fffff. Tan: its seven res 9 children.
Orange: res 11 cells round an approximate point in Llandough village; dark orange = 8b19586b3923fff.



North up. Approximate point, within about 300 m. Parcel outlines not drawn. Cells from the H3 library 4.x.

Download Links

See below a table of the download links for the Annexed items in this document:

Download Link of Annexures

Date	Description	PDF	SVG	PNG
2026-11-01	H3 resolutions 8 9 and 11 over the Cedwyn pilot area (schematic)			cedwyn-h3-layers.png

Document Author(s):

DATRO Consortium